but only and when the patient can be expected to be adherent to wearing the shoes. (Weak; Low) IWGDF 2015 (Adopted)

4.5.4 Instruct an at-risk patient with diabetes to wear properly fitting footwear to prevent a first foot ulcer, either plantar or non-plantar, or a recurrent non-plantar ulcer. When a foot deformity or a pre-ulcerative sign is present, consider prescribing therapeutic shoes, custom-made insoles, or toe orthosis*. (Strong; Low) IWGDF 2015 (Adopted)
*Toe orthosis:- An in-shoe orthosis to achieve some alteration in the function of the toe.

4.5.5 To prevent a recurrent plantar foot ulcer in an at-risk patient with diabetes, prescribe therapeutic footwear that has a demonstrated plantar pressure relieving effect during walking and encourage the patient to wear this footwear. IWGDF 2015 (Adapted)

4.5.6 Instruct a patient with diabetes not to use conventional or standard therapeutic footwear to heal a plantar foot ulcer. IWGDF 2015 (Adapted)
Explanatory note: Use footwear with following features -
Sandals: should have adjustable straps, insole, full heel counter and rigid outsole.
Shoes: should have wide toe box extra depth and without laces.

4.5.7 Consider using shoe modifications, temporary footwear, toe spacers or orthoses to offload and heal a non-plantar foot ulcer without ischemia or uncontrolled infection in a patient with diabetes. The specific modality will depend on the type and location of the foot ulcer. (Weak; Low) IWGDF 2015 (Adopted)

4.5.8 If other forms of biomechanical relief are not available, consider using felted foam* in combination with appropriate footwear to offload and heal a neuropathic foot ulcer without ischemia or uncontrolled infection in a patient with diabetes. (Weak; Low) IWGDF 2015 (Adopted)
Felted foam - A fibrous, unwoven material backed by foam with absorbing and cushioning characteristics. The foam is generally 'rubber foam' or 'PU foam' which is formed by either a polyester or polyether polyol resin, in conjunction with water and Toluene di Isocyanate, along with various catalysts and blowing agents and colouring pigments to give the desired compression.

4.6: Diabetic foot with osteomyelitis and CHARCOT'S FOOT

4.6 Treatment of Diabetic foot with osteomyelitis

4.6.1 For an infected open wound, perform a probe-to-bone test; in a patient at low risk for osteomyelitis a negative test largely rules out the diagnosis, while in a high risk patient a positive test is largely diagnostic. (Strong; High)IDSA 2012 guidelines (Adopted)

4.6.2 Markedly elevated ESR is suggestive of osteomyelitis in suspected cases. IWGDF 2015 (Adapted)
Explanatory note: Tests for serum inflammatory markers are costly and not widely available, except ESR. Also these tests are not diagnostic of DFO.

4.6.3If osteomyelitis is suspected in a person with diabetes but is not confirmed by initial X-ray, consider an MRI to confirm the diagnosis. In places where MRI is unavailable, diagnose osteomyelitis by the PTB test (clinically) and/or taking a Bone biopsy and culture. (Adapted from NICE clinical guideline NG19 and Expert group discussion)
Explanatory note: Expert Consensus says that as availability of MRI is limited across the country, it is recommended to use MRI wherever available. At the primary and secondary health centre levels, the PTB test and bone biopsy and culture are more feasible and economical and reasonably accurate.

4.6.4 A definite diagnosis of bone infection usually requires positive results on both histological and microbiological examinations of an aseptically obtained bone sample, but this is usually required only when the diagnosis is in doubt or determining the causative pathogen's antibiotic susceptibility is crucial. (Strong; Moderate) IDSA 2012 guidelines (Adopted)

4.6.5 Avoid using results of soft tissue or sinus tract specimens for selecting antibiotic therapy for osteomyelitis as they do not accurately reflect bone culture results. (Strong; Moderate) IDSA 2012 guidelines (Adopted)

4.6.6 When a radical resection leaves no remaining infected tissue*, we suggest prescribing antibiotic therapy for only a short duration (2–5 days). When there is persistent infected or necrotic bone, we suggest prolonged (≥4 weeks) antibiotic treatment. (weak, low) IDSA 2012 guidelines (Adapted)
Explanatory note: *A proximal bone histopath to be done if available to get a clear margin and confirm that no infected bone remains.

4.6.7 For specifically treating Diabetic foot osteomyelitis, we do not currently support using adjunctive treatments such as hyperbaric oxygen therapy, growth factors (including granulocyte colony stimulating factor), maggots (larvae), or topical negative pressure therapy (eg, vacuum-assisted closure). (weak, low) IDSA 2012 guidelines (Adapted)
Also refer to Table 4 and 5 in the Annexure below, for explanatory notes.

4.7. Charcot's Foot
4.7.1 Suspect acute Charcot arthropathy if there is redness, warmth, swelling or deformity (in particular, when the skin is intact), especially in the presence of peripheral neuropathy or renal failure. Think about acute Charcot arthropathy even when deformity is not present or pain is not reported. NICE clinical guideline NG19 (Adopted)

4.7.2 Refer the person with suspected charcot's foot early (within one week) to the - Diabetic Foot care centre to confirm the diagnosis, and offer non-weight-bearing treatment until definitive treatment can be started. NICE clinical guideline NG19(Adapted)
Diabetic Foot care centre:- In India, since there are no minimum standards of services offered to the diabetic foot patients, in our recommendations we have used this term to denote this facility, which may exist at the General Practitioner's office, Primary health centre, Secondary care centre or at a tertiary care centre. Preferably, the diabetic foot care centre should consist of atleast a surgeon, a physician, and an orthotist.

4.7.3 If acute Charcot arthropathy is suspected, X-ray the affected foot. Consider an MRI if the X-ray is normal but clinical suspicion still remains. NICE clinical guideline NG19 (Adopted)

4.7.4 Distinguishing the bony changes of osteomyelitis from those of the less common entity of diabetic neuro-osteoarthropathy (Charcot foot) may be particularly challenging and requires considering clinical information in conjunction with imaging. IDSA 2012 guidelines (Adopted)

4.7.5 Clinical clues supporting neuro-osteoarthropathy in this context include midfoot location and absence of a soft tissue wound, whereas those favoring osteomyelitis include presence of an overlying ulcer (especially of the forefoot or heel), either alone or superimposed on Charcot changes. IDSA 2012 guidelines (Adopted)